UCIS

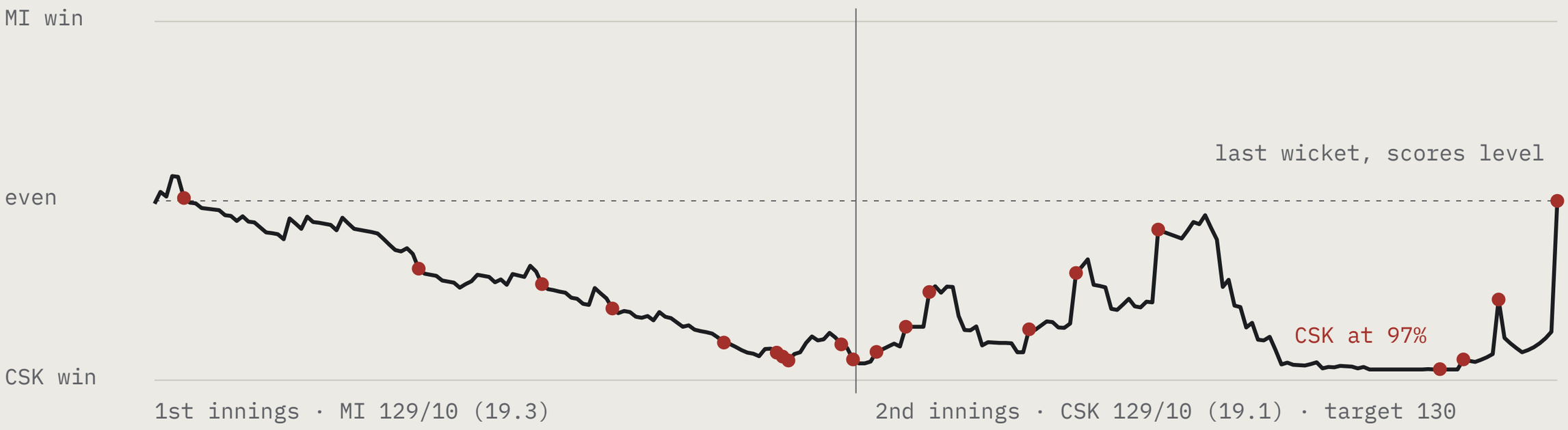
Unified Cricket Intelligence System

A battery-powered cricket box. One Arduino UNO Q fetches live IPL scores, plays out whole matches ball by ball and serves a phone dashboard. A 2.8-inch touch screen keeps the score at the table.

CSK DC GT KKR LSG MI PBKS RR RCB SRH

One simulated match, ball by ball

Mumbai Indians v Chennai Super Kings, generated on the UNO Q by the hub's simulator (seed 25). The line is the win chance the box shows after every delivery. MI made 129 all out in 19.3 overs. CSK needed 130 and were 129 all out with 5 balls left: **a tie.**



● wicket line: chance MI win after each ball (a wide counts as a step)

MI batting

1	1	**4**	·	**6**	1	W	
2	·	1	·	1	1	1	
3	·	1	·	wd	·	1	·
4	·	1	1	·	**6**	·	
5	·	**4**	·	1	1	1	
6	·	**4**	·	·	1	1	
7	1	1	·	·	·	1	
8	2	·	W	·	1	1	
9	·	1	1	·	wd	2	3
10	1	1	·	2	·	**4**	
11	·	1	**4**	·	W	·	
12	1	1	1	·	1	·	
13	1	**6**	·	·	W	·	
14	2	1	·	1	2	·	
15	**4**	·	1	·	·	2	
16	·	1	1	1	·	W	
17	1	·	·	·	1	·	
18	**6**	2	W	W	W	**6**	
19	wd	**6**	**4**	·	2	**4**	·
20	W	·	W				

CSK chasing

1	·	1	·	W	·	·	
2	·	wd	W	1	1	1	W
3	·	2	·	1	wd	**4**	1
4	·	**6**	·	·	1	1	
5	1	**4**	1	W	1	·	
6	·	1	2	·	·	W	
7	·	·	**4**	1	1	**4**	
8	1	·	·	2	1	·	
9	1	W	1	1	1	1	
10	·	·	1	·	2	2	
11	wd	·	**4**	1	**4**	·	**4**
12	1	·	**4**	**6**	·	wd	1
13	·	·	·	**4**	·	1	
14	·	1	1	wd	·	**4**	1
15	·	2	·	·	1	1	
16	1	1	·	**4**	W	1	
17	1	1	W	1	1	·	
18	·	·	W	**4**	1	1	
19	1	·	·	·	·	·	
20	W						

Hardware

Arduino UNO Q runs the hub, simulator, win model and dashboard on battery. No laptop at match time.

ESP32 “Cheap Yellow Display”, 2.8-inch touch screen, is the scoreboard. It talks to the UNO Q over three wires:

```
UNO Q D1  →  IO35  blue
UNO Q D0  ←  IO22  yellow
GND       —  GND   black
```

UART 115200 8N1 · CYD port P3

Two screens

Live. The current IPL match from CricketData.org. With no live game it says so and shows the next fixture.

Simulation. A full T20 played out ball by ball, with the win-chance line drawn above.

Tap the screen to switch. Hold it for a QR code that opens the dashboard.

Dashboard

The UNO Q runs its own Wi-Fi hotspot. Phones join it and open the dashboard: no app store, no internet needed for the simulation.

Pages: overview, live centre, scoring, simulation, analytics, teams, tournaments and device control.

FastAPI hub in Docker on the board · React front end

How the win chance is worked out

Wickets in hand w give a run capacity $z = 20 \cdot w^{1.48}$. With b balls left the side can expect $z(1 - e^{-1.6b/z})$ more runs. In a chase, expected runs minus runs needed goes through a logistic curve scaled by $\sigma = 2.5 + 1.6\sqrt{b}$, held between 3% and 97%. The CYD (C++) and the hub (Python) run the same code, so both show the same number.

Data path

```
CricketData.org
  ↓ hub on UNO Q
  ↓ UART → CYD screen
  ↓ Wi-Fi → phones
```

Try it: join Wi-Fi **UCIS-Cricket**, then scan the code or open 10.42.0.1:8000.

github.com/Wired-Navi0x17/IOT-SEE · live data: CricketData.org · player ratings are placeholders until the 2026 squads are loaded